08 Gas phase concentrations

Why are we learning this?

In the previous lecture, we learned how conversion affects the number of moles present in a reactor. Unfortunately, most gas-phase reactor design problems are more complicated because concentration depends not only on the number of moles present, but also on:

- Temperature
- Pressure
- Total volume

In this lesson, we will develop concentration expressions that remain valid when gas-phase reactors experience:

- Conversion
- Expansion or contraction
- Temperature changes
- Pressure changes

By the end of today's lesson, you should understand:

- How ideal gas relationships determine concentration.
- Why gas-phase concentrations are more complicated than liquid-phase concentrations.
- How conversion influences the total molar flow rate.
- The physical meaning of δ and ε (which are simply bookkeeping terms).
- How to express gas-phase concentrations in terms of conversion, temperature, and pressure.
- Why these relationships are essential for reactor design.

Course Roadmap

Introduction to CRE
↓

General Balance Equation
↓
Common Reactor Models
(CSTR, Batch, PFR)
↓
►Reaction Progress◄
(Conversion)
↓
Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)
↓
Rate Laws
(Kinetics)
↓
Equilibrium
(Reversible Reactions)
↓
Equilibrium Conversion
(Thermodynamic Limitations)
↓
One Reaction: Solving in terms of Reaction Progress
↓
One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)
↓
Catalytic Reactions
↓
Numerical Methods
(Python)
↓
Energy Balances
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One Reaction: Non-Isothermal Reactor Design
↓
Multiple Reactions
↓
Reactor Safety

↓
Challenging the Well-Mixed Assumption
(Residence Time Distributions)
↓
Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Big Idea

For gases: Concentration depends on

Moles
+
Pressure
+
Temperature

All three can change inside a reactor.

The concentrations of ideal gases can be calculated using the

$$C = \frac{N}{V} =$$

For a flow reactor, $C =$ $= \frac{P}{RT}$

Why Are Gas-Phase Reactions Harder?

In many liquid-phase systems, volume can be treated as constant. Gas-phase systems are different.

As a reaction proceeds:

- Total moles may change.
- Temperature may change.
- Pressure may change.

All three influence gas phase concentration but rarely liquid phase concentration. As a result, gas-phase reactor design usually requires more bookkeeping than liquid-phase reactor design.

At the entrance to a flow reactor, $C_{T0} = \frac{F_{T0}}{\upsilon_0} = \frac{P_{T0}}{RT_0}$

Where 'T' stands for

For an ideal gas,

Total Concentration vs Species Concentration

The subscript **T** refers to the total mixture.

Examples:

CA = concentration of A

CB = concentration of B

CT = total concentration

Total concentration includes contributions from every species present in the reactor.

Consider a PFR carrying out the reaction A + B --> C:

The Reactor Design Problem

Most rate laws require concentrations. For example:

-rA = k*CACB

To use this rate law inside a reactor design equation, we need concentrations expressed in terms of variables we are actually solving for. Today, that variable is conversion.

This PFR is non-isothermal and experiences pressure drop. How do we compute C_A, C_B, and C_C as functions of conversion?

$$C_j = \frac{F_j}{\upsilon}$$

$$F_j =$$

To get υ, we use "PVT" laws:

$$\upsilon = \frac{F_{T0}RT}{P}$$

Hence, $\frac{\upsilon}{\upsilon_0} =$

Hence, at any point in the PFR, $\upsilon = \upsilon_0 \left(\frac{F_T}{F_{T0}}\right)\left(\frac{P_0}{P}\right)\left(\frac{T}{T_0}\right)$

$$C_j = \frac{F_j}{\upsilon} =$$

Hence, $C_j = \left(\frac{F_{T0}}{\upsilon_0}\right)\left(\frac{F_j}{F_T}\right)\left(\frac{P}{P_0}\right)\left(\frac{T_0}{T}\right) =$ $\left(\frac{F_j}{F_T}\right)\left(\frac{P}{P_0}\right)\left(\frac{T_0}{T}\right)$

Why Do Pressure and Temperature Appear?

For ideal gases:

$$C = \frac{P}{RT}$$

Therefore:

- Increasing pressure increases concentration.
- Increasing temperature decreases concentration.

Even if conversion stays fixed, changes in pressure and temperature still alter concentration. This is why both terms appear in the final equation.

Why Does Total Flow Rate Matter?

The volumetric flow rate of an ideal gas depends on:

Total molar flow rate
+
Temperature
+
Pressure

If any of these quantities change, the volumetric flow rate changes.

Since concentration equals: Molar Flow Rate divided by Volumetric Flow Rate

changes in molar flow rate directly influence concentration.

In terms of conversion:

Define $\delta = \frac{\text{change in the total number of moles}}{\text{mole of LR reacted}}$

For example, for the reaction aA + bB --> cC + dD, where species A is the limiting reactant: $\delta = \frac{c+d-a-b}{a}$

For example, for the NO oxidation reaction: NO + 1/2 O_2 --> NO_2, $\delta =$

Then, $F_T = F_{T0} + F_{LR,0}\delta X$

Then $\frac{F_T}{F_{T0}} =$

Define $\epsilon = y_{LR,0}\delta$, where $y_{LR,0}$ is

Then, $\upsilon = \upsilon_0 \left(\frac{F_T}{F_{T0}}\right)\left(\frac{P_0}{P}\right)\left(\frac{T}{T_0}\right) = \upsilon_0(\qquad\qquad\qquad\qquad) \left(\frac{P_0}{P}\right)\left(\frac{T}{T_0}\right)$

We can do some algebra to arrive at $C_j = \frac{C_{LR,0}(\Theta_j + \nu_j X)}{1+\epsilon X}\left(\frac{P}{P_0}\right)\left(\frac{T_0}{T}\right)$

Physical Meaning of δ

δ measures how many total moles are created or destroyed when one mole of limiting reactant reacts.

Examples

A → 2B: Total moles increase: δ > 0

2A → B: Total moles decrease: δ < 0

A → B: Total moles remain unchanged: δ = 0

δ determines whether the reacting gas mixture expands or contracts.

Physical Meaning of ε

ε accounts for two effects simultaneously:

1. How much the total number of moles changes (δ).
2. How important the limiting reactant is in the feed.

Large values of ε correspond to larger changes in volumetric flow rate as conversion increases. Think of ε as a convenient way to track the extent of gas expansion or contraction during reaction.

Reading the Final Concentration Equation

Each factor has a physical meaning:

($\Theta_j + \nu_j X$): Tracks how stoichiometry changes the amount of species j.

$1 + \varepsilon X$: Accounts for expansion or contraction of the gas mixture.

P/P_0: Accounts for pressure changes.

T_0/T: Accounts for temperature changes.

Together, these terms allow concentration to be written entirely in terms of reactor variables.

Summary and Key Takeaways

Today we developed concentration expressions for reacting gas-phase systems.

Key Ideas

- ☑ Gas-phase concentrations depend on moles, pressure, and temperature.
- ☑ Conversion changes the number of moles present in the reactor.
- ☑ The total molar flow rate influences volumetric flow rate.
- ☑ δ measures the change in total moles per mole of limiting reactant reacted.
- ☑ ε captures the extent to which the change in the total moles influences gas expansion or contraction.

✓ Pressure and temperature effects must be included when modeling gas-phase reactor concentrations.

Most Important Idea

Concentration changes because:

Reaction changes moles
+
Pressure changes density
+
Temperature changes density

The final concentration equation combines all three effects into a form that can be used directly in reactor design calculations.